

MOTOR CITY FACILITATOR TRAINING AND CLASSROOM SOP

Audience: professors, instructors, teaching assistants, course coordinators, and backup facilitators

Website: <https://motorcity.boeingcenter.com>

Purpose: make a new facilitator independently ready to prepare, run, recover, finish, debrief, and export a Motor City class session.

Document owner: course owner

Review cycle: before the first class each term and after any major interface change

Last verified: August 27, 2026

This is the training and operating standard. Keep the [one-page checklist](#) beside you during class, use the [full facilitator manual](#) for control-by-control detail, and print the [large-type classroom guide](#) for a podium copy. A standalone [printable SOP](#) is available for training and sign-off.

What qualified means

A qualified facilitator can, without coaching:

1. Create and verify the intended setup before students join.
2. Explain the join code, recovery code, final round, WIP round, and timer correctly.
3. Run the lobby, projector, factory floor, leaderboard, and private controls without exposing scoring information early.
4. Recover a facilitator or student session and remove an accidental participant.
5. End at the intended scoring point, conduct the WIP reveal, and explain the result.
6. Distinguish a **Proven optimal simulation** from a legal **Best run found**.
7. Download and interpret the Excel workbook before the 12-hour retention window ends.

Qualification path

Allow 45 minutes for a first-time facilitator:

Time	Activity	Evidence of completion
5 min	Read QUICK_START.md	Can describe the class flow from memory
20 min	Complete the rehearsal below	Runs both facilitator and test-student views
10 min	Review the incident and debrief tables	Chooses the correct response in each scenario
10 min	Complete the sign-off	Another facilitator observes the required tasks

Annual users should repeat the rehearsal before their first class of the term. A backup facilitator should qualify before receiving a live recovery code.

Roles

One person may hold more than one role, but name the owner before class.

Role	Owns
Lead facilitator	Setup, start/end decisions, private scoring, reveal, and final export
Classroom support	Student joining, recovery, duplicate removal, and device questions
Course owner	Learning objectives, approved economics/scoring changes, and data handling
Technical contact	Website outage or server issue that the classroom procedure cannot resolve

Only the lead facilitator or an explicitly designated backup should receive the facilitator recovery code. Students receive the six-character join code, never the facilitator recovery code.

Standard teaching configuration

Unless the course owner has approved a different exercise, use the preloaded **Recommended default**:

- all four car models
- exact **Original 25-round Team Sequence** from v1
- standard revenue and WIP economics
- final round 25 and WIP penalty round 25
- 10 minutes per round for rounds 1-10
- 5 minutes per round for rounds 11-25

The default **Optimal Run** is a proven \$82.00 reference: 28 shipped cars, consisting of 24 blue and 4 red, with zero final WIP. It is a facilitator reference, not a student target that must be shown before play.

Non-negotiable operating rules

1. Keep the facilitator recovery code private.
 2. Verify **Private scoring**, then collapse it whenever students can see the admin screen.
 3. Wait for **synced** after a connection interruption; do not repeat commands rapidly.
 4. Confirm the roster, final round, WIP round, and timer before **Start production**.
 5. Confirm the two scoring rounds again before **End production**.
 6. Preserve the reveal: live standings are revenue only until unfinished cars are charged.
 7. Call a result optimal only when it says **Proven optimal simulation**.
 8. Download the workbook before the session is 12 hours old.
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Required rehearsal

Use a temporary short session so the whole lifecycle fits in 20 minutes.

Prepare the rehearsal

1. Open the website and choose **Facilitate**.
2. Enter `Training - your name` as the facilitator name.
3. Choose **Classic 10-round sequence**, leave only the green model active, set final and WIP rounds to 2, and turn the timer off.
4. Create the session and save the facilitator recovery code privately.
5. Open a private/incognito window so the test student has separate browser credentials from the facilitator. Join there as `Test Student` with the six-character join code.

Run the rehearsal

1. Confirm the test student appears in **Roster**, then open and collapse **Private scoring**.
2. Press **Present** and verify that the projector view shows only public information.
3. Press **Start production**.
4. As the test student, make at least one legal move or allocation and advance a round.
5. Close the student tab, choose **Rejoin**, and restore the same board with the student recovery code.
6. From **Factory floor**, issue **New code for Test Student** and confirm the old recovery code no longer works.
7. Press **Calculate reference run**. Explain the waiting/calculating state and the final proof label, then open **View round history**.
8. Open the student's history from the leaderboard and identify issued, exchanged, and unused resources.
9. End production, press **Charge the unfinished cars** in the presenter, and explain why revenue order can differ from final-score order.
10. Download the Excel workbook and locate **Game Details**, **Player Overview**, **Game Stats**, and the student sheet.

Rehearsal pass criteria

The facilitator passes when they complete the lifecycle without exposing the facilitator recovery code, can explain both scoring rounds and both optimizer labels, recover the test student, preserve the reveal, and produce the workbook. Repeat only the missed tasks before sign-off.

Classroom SOP

T-24 hours to T-30 minutes: prepare

1. Confirm the website opens on the instructor computer and the projector computer.
2. Confirm the course owner has chosen the standard setup or documented every customization.
3. Decide whether students should enter an optional ID/email for gradebook matching.
4. Name the lead facilitator, classroom support person, and technical contact.
5. Have the PDF or [QUICK_START.md](#) available offline.
6. Do not create the live session more than 12 hours before it will finish.

T-15 minutes: create and verify

1. Choose **Facilitate**, enter the facilitator name, and review the setup.
2. If the setup was changed unintentionally, press **Restore**.
3. Create the session and save the recovery code in an approved private location.
4. For a custom setup, start **Calculate reference run** now; it may take several minutes.
5. Open a second browser window for the projector. Choose **Rejoin** with the same facilitator name and recovery code, then press **Present**.
6. On the private instructor screen, verify player count, final round, timer, and **Private scoring**. For the recommended default, the timer is already enabled at 10 minutes for rounds 1-10 and 5 minutes for rounds 11-25. Collapse private scoring afterward.

The presenter window shows public join information before play and revenue-only standings during play. Opening **Private scoring** on the separate instructor window does not add it to **Present**.

T-5 minutes: admit students

1. Project the website and join code.
2. Ask students to choose **Join**, enter their name, and enter an ID only if the course requires one.
3. Watch **Roster**. Resolve duplicate or incorrect names before starting.
4. Tell students: "Your factory is independent. Wait for the instructor before starting. If your device closes, use Rejoin rather than creating another player."

Start

Use this final verbal check:

I have the expected roster. The final round, WIP round, and timer match today's plan. Private scoring is collapsed. The projector shows public information only.

Press **Start production** once. Confirm student boards open and the facilitator status returns to **synced**.

During play

- Use **Factory floor** to scan round progress, paint occupancy, unused resources, and quiet boards.
- Use **Leaderboard** for revenue standings. Do not describe it as final score during play.
- Select a student name when you need evidence from their round history.
- Intervene on access or mechanics; do not prescribe the optimal production mix unless that is an explicit learning activity.
- Keep the instructor controls off the projector. Use **Present** for the room view.
- Note the rounds where WIP rises, throughput stalls, or unused resources accumulate. Those are the best debrief examples.

End and reveal

1. Announce the last playable round before ending.
2. Open **Private scoring** and verify **Score up to round** and **WIP penalty round**.
3. Press **End production**, then **Yes, end it**. Do not repeat the action while syncing.
4. Let the room observe the revenue order first.
5. Switch to the presenter window. After ending is confirmed, **Charge the unfinished cars** appears there; press it when you are ready.
6. Pause for the reordered standings before explaining them.
7. Press **Show it again** only when a second look helps the discussion.

Export and close

1. Download **Excel workbook**; use CSV only when a flat table is specifically needed.
2. Open the workbook and verify the expected student count and identifiers.
3. Store the file according to the course's student-data policy.
4. Record the join code, class date, setup name, and any incident in course notes. Do not record recovery codes in shared notes.
5. Close the session after the export is confirmed. Server data expires 12 hours after session creation, regardless of when production ends, so export immediately after the run.

Intervention decision table

Situation	Immediate action	Do not do
Student tab or device closed	Student chooses Rejoin with their own recovery code	Do not create a duplicate player
Student lost recovery code	Use New code for [name] and read it privately	Do not post the code to the room
Wrong or duplicate participant	Use Remove , then confirm the correct name	Do not remove while unsure which board is active
Facilitator tab closed	Rejoin with facilitator name and facilitator recovery code	Do not join as a student
Screen says offline	Treat it as a temporary connection/sync state: pause actions and wait for synced . If it does not return, use the outage response below	Do not click the same command repeatedly
Student timer reaches zero	Have the student press Advance round in the locked-board message	Do not expect an automatic round change
Class must end early	Set both scoring fields to valid reached rounds, verify them, then end	Do not leave WIP later than the score cutoff
Reference is still calculating	Continue teaching; check it later	Do not delay gameplay solely for the reference
Reference says best found	Treat it as legal, not proven optimal	Do not label it "the optimum"
Reference is unavailable	Retry once; continue without it if it still fails	Do not restart the live class
Projector shows private controls	Close presenter/admin view immediately and restore Present	Do not discuss the hidden value before the reveal
Website is unavailable for everyone	Pause, record time and symptoms, contact the technical owner	Do not switch systems mid-run without an approved fallback

Expert debrief

The debrief should connect visible behavior to supply-chain principles, not merely announce a winner.

Ten-minute sequence

1. **Observe:** "What do you notice about the revenue standings before WIP is charged?"
2. **Reveal:** charge unfinished cars and allow the class to react to rank changes.
3. **Explain:** distinguish throughput, revenue, WIP, penalty, and final score.
4. **Diagnose:** open one useful student history. Find the round where WIP accumulated or flow stalled.
5. **Compare:** open **Optimal Run** history. Compare decision patterns, not only totals.
6. **Transfer:** ask what policy the team would change in a real production system.

Useful questions

- Where did inventory wait, and what made that station a bottleneck?
- Did resources create throughput or merely create more WIP?
- Which lane-priority or conversion decision changed later options?
- Why can the highest-revenue player lose after the WIP charge?
- What does a smooth flow look like in the round history?
- What can a proven reference establish, and what can it not teach about human coordination?

Use the reference simulation as evidence, not as a script students were expected to discover. Multiple legal histories may attain the same score.

Secure handoff to a backup facilitator

Use an approved private channel and hand off:

- website and join code
- facilitator name and recovery code
- current session state: waiting, active, or finished
- expected roster and any duplicate/removed participant
- resource plan, models, economics, final round, WIP round, and timer
- last confirmed action and any unresolved incident
- who owns the final reveal and export

The receiver should rejoin on their own device and confirm **synced** before the original facilitator steps away. Never place a facilitator recovery code in slides, chat, shared course notes, or the exported workbook.

Facilitator sign-off

Observer initials each item. All items are required for independent qualification.

Competency	Initials
Identifies the standard 25-round setup and restores it after a change	
Explains final round versus WIP penalty round	
Configures and validates contiguous timer ranges	
Protects and uses the facilitator recovery code	
Runs join, roster verification, start, presenter, and end	
Keeps private scoring out of the student/projector view	
Recovers a student and rotates a lost recovery code	
Removes a duplicate participant using confirmation	
Responds correctly to offline and timer-expired states	
Interprets proven optimal versus best found	
Navigates student and reference round histories	
Conducts the revenue-to-WIP reveal and debrief	
Downloads and verifies the Excel workbook	
Completes a secure backup-facilitator handoff	

Facilitator: _____ **Observer:** _____ **Date:** _____

Requalify after a major interface change or when any required task can no longer be completed without assistance.